

1) Implement an MPI program using N processes to find $1! + 2! + \dots + N!$. Use scan. Also, handle different errors using error handling routines.

```
5CSED1@dsl-11:~/Documents/240905318_Anirvin_PCAP/week6$ mpicc pgm1.c -o pgm1
5CSED1@dsl-11:~/Documents/240905318_Anirvin_PCAP/week6$ mpirun -np 6 ./pgm1
Process 1: factorial = 2, sum = 3
Process 2: factorial = 6, sum = 9
Process 3: factorial = 24, sum = 33
Process 4: factorial = 120, sum = 153
Process 0: factorial = 1, sum = 1
Process 5: factorial = 720, sum = 873
```

```
5CSED1@dsl-11:~/Documents/240905318_Anirvin_PCAP/week6$ mpicc pgm1.c -o pgm1
5CSED1@dsl-11:~/Documents/240905318_Anirvin_PCAP/week6$ mpirun -np -1 ./pgm1
Process 0: Invalid rank, error stack:
internal_Send(120): MPI_Send(buf=0x7ffcf39a0750, count=1, MPI_INT, 1, 0, MPI_COMM_WORLD) failed
internal_Send(78): Invalid rank has value 1 but must be nonnegative and less than 1
```

2) Implement an MPI program to read a 3 X 3 matrix. Enter an element to be searched in the root process. Find the number of occurrences of this element in the matrix using three processes.

```
5CSED1@dsl-11:~/Documents/240905318_Anirvin_PCAP/week6$ mpirun -np 5 ./pgm2
Enter 3 X 3 matrix:
1
1
4
3
7
4
9
9
10
Enter element to search: 4
Number of occurrences = 2
```

3) Implement an MPI program to read 4 X 4 matrix and display the following output using four processes.

I/p matrix:

```
1 2 3 4
1 2 3 1
1 1 1 1
2 1 2 1
```

O/p matrix:

```
1 2 3 4
2 4 6 5
3 5 7 6
5 6 9 7
```

```

5CSED1@dsl-11:~/Documents/240905318_Anirvin_PCAP/week6$ mpicc pgm3.c -o pgm3
5CSED1@dsl-11:~/Documents/240905318_Anirvin_PCAP/week6$ mpirun -np 4 ./pgm3
Enter 4 X 4 matrix:
1
2
3
4
1
2
3
1
1
1
1
1
2
1
2
1
Process 0: 1 2 3 4
Process 1: 2 4 6 5
Process 2: 3 5 7 6
Process 3: 5 6 9 7

```

4) Implement parallel odd even transposition sort

```

5CSED1@dsl-11:~/Documents/240905318_Anirvin_PCAP/week6$ mpirun -np 10 ./pgm4
Enter 10 values:
37 12 45 8 23 91 4 56 19 72
Process 0: 4
Process 1: 8
Process 2: 12
Process 3: 19
Process 4: 23
Process 5: 37
Process 6: 45
Process 7: 56
Process 8: 72
Process 9: 91

```